PROBLEM 1.9

Partial Fractions • Carrying Capacity • Malthusian Limit

Logistic Population Growth Model (Verhulst)

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A refinement over Malthus' population growth model discussed in Chapter 1 is provided by the logistic equation:

$$\frac{dp}{dt} = kp\left(1 - \frac{p}{M}\right)$$

where $p(t)$ is the population size at time t, k is its growth rate (difference between birth rate and death rate), and M is the maximum sustainable population (carrying capacity).

Solve the equation for a generic initial condition $p(t_0) = p_0$ and discuss the evolution of the population for $t \to \infty$, comparing with Malthus' ($M \to \infty$) prediction.

PHASE 2: FUNDAMENTAL FORMULATION

First-order nonlinear autonomous ODE. Separate variables by moving all terms in p to the LHS, decompose into partial fractions, and integrate from (t_0, p_0) to (t, p).

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: Separation of Variables:

$$\frac{dp}{dt} = \frac{k}{M}p(M-p) \implies \frac{dp}{p(M-p)} = \frac{k}{M}\,dt \qquad (1.9.1)$$

Step 2: Partial Fraction Decomposition:

$$\frac{1}{p(M-p)} = \frac{A}{p} + \frac{B}{M-p} \implies 1 = A(M-p) + Bp$$

At $p = 0 \implies A = \frac{1}{M}$; at $p = M \implies B = \frac{1}{M}$. Substituting back:

$$\frac{1}{M}\left(\frac{1}{p} + \frac{1}{M-p}\right) dp = \frac{k}{M}\, dt \implies \left(\frac{1}{p} + \frac{1}{M-p}\right) dp = k\, dt \tag{1.9.2}$$

Step 3: Definite Integration:

$$\int_{p_0}^{p} \left(\frac{1}{u} + \frac{1}{M-u}\right) du = \int_{t_0}^{t} k\, ds$$

Recall $\int \frac{1}{M-u} du = -\ln|M-u|$:

$$\left[\ln|u| - \ln|M-u|\right]_{p_0}^{p} = k(t-t_0) \implies \left[\ln\left|\frac{u}{M-u}\right|\right]_{p_0}^{p} = k(t-t_0)$$

$$\ln\left|\frac{p(M-p_0)}{(M-p)p_0}\right| = k(t-t_0)$$

Step 4: Exponentiation and Isolation of $p(t)$:

$$\frac{p(M-p_0)}{(M-p)p_0} = e^{k(t-t_0)} \implies p(M-p_0) = p_0(M-p)e^{k(t-t_0)}$$

$$p(M-p_0) + pp_0e^{k(t-t_0)} = Mp_0e^{k(t-t_0)} \implies p\left[(M-p_0) + p_0e^{k(t-t_0)}\right] = Mp_0e^{k(t-t_0)}$$

Dividing numerator and denominator by $e^{k(t-t_0)}$:

$$\mathbf{p(t) = \frac{Mp_0}{p_0 + (M-p_0)e^{-k(t-t_0)}}} \tag{1.9.3}$$

Step 5: Asymptotic Limits:

- As $t \to \infty$: $\lim_{t\to\infty} e^{-k(t-t_0)} = 0 \implies \lim_{t\to\infty} p(t) = \frac{Mp_0}{p_0+0} = \mathbf{M}$.

- As $M \to \infty$ (Recovery of Malthus): Dividing numerator and denominator of (1.9.3) by M:

$$p(t) = \frac{p_0}{\frac{p_0}{M} + \left(1 - \frac{p_0}{M}\right) e^{-k(t-t_0)}} \implies \lim_{M\to\infty} p(t) = \frac{p_0}{0 + (1-0)e^{-k(t-t_0)}} = \mathbf{p_0 e^{k(t-t_0)}}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$p(t) = \frac{Mp_0}{p_0 + (M - p_0)e^{-k(t-t_0)}}$$

$$\lim_{t\to\infty} p(t) = M, \quad \lim_{M\to\infty} p(t) = p_0 e^{k(t-t_0)}$$

Physical Interpretation: The logistic curve exhibits a sigmoidal shape when $p_0 < M/2$, passing through an inflection point of maximum growth rate at $p = M/2$.